MS 6015: Multivariate Statistical Methods for Business

Classification trees

March 30, 2026

Learning Objectives

1. Understand foundations of decision trees (also known as decision tree learning and classification trees) and how they are used for solving classification problems.
2. Learn how to construct tree techniques such as Chi-square Automatic Interaction Detection (CHAID) and Classification and Regression Tree (CART) and their application in solving classification problems.
3. Understand different splitting strategies such as Chi-square Test, F-test, Gini Impurity Index and entropy used in decision trees.
4. Understand how to generate business rules using decision trees.

Outline

Decision Trees

- Decision trees (also known as decision tree learning or classification trees) are a collection of predictive analytics techniques that use tree-like graphs for predicting the value of a response variable (or target variable) based on the values of explanatory variables (or predictors).
- They are effective for solving classification problems in which the response variable (target variable) takes discrete values and regression problems in which the outcome variable takes continuous values.

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Tree Development Criteria

Decision trees use the following criteria to develop the tree:

1. Splitting Criteria: Splitting criteria are used to split a node (set of data) into subsets.
2. Merging Criteria: When the predictor variable is categorical with *n* categories, it is possible that not all *n* categories may be statistically significant.
3. Stopping Criteria: Stopping criteria is used for pruning the tree (stopping the tree from further branching) to reduce the complexity associated with business rules generated from the tree.

Generating Decision Trees

Following steps are used for generating decision trees:

1. Start with the root node in which all the data is present.
2. Decide on a splitting criterion and stopping criteria: Root node split into two or more subsets tree branches (called edges) using the splitting criterion.
3. The terminal nodes (aka leaf nodes) will not have any outgoing edges.
4. Terminal nodes are used for generating business rules (prediction rules) which can be further used for predicting the value of the outcome variable.
5. Tree pruning (a process for restricting the size of the tree) is used to avoid large trees and over-fitting the data.

Popular decision tree techniques

There are many decision tree techniques which differ in the strategy, the two most popular decision tree techniques

1. Chi-square Automatic Interaction Detection (CHAID)
2. Classification and Regression Trees (CART)

will be discussed.

Chi-square Automatic Interaction Detection (CHAID)

CHAID (Kass, 1980) is an extension of Automatic Interaction Detection (AID), which is designed to categorize the dependent variable using categorical predictors.

- CHAID partitions the data into mutually exclusive, exhaustive, subsets that best describe the dependent categorical variable.
- CHAID is an iterative procedure that examines the predictors (or classification variables) and uses them in the order of their statistical significance.

Splitting Rules

The splitting in CHAID is based on the type of the dependent variable (or target variable).

- Chi-square Test of Independence when the response variable, Y, is discrete.
- F-test when the response variable, Y, is continuous.
- Likelihood Ratio Test when the response variable, Y, is ordinal.

Example: German credit

- German Credit Rating Data
- Dependent variable: Categorical ($Y = 0$ good credit, $Y = 1$ bad credit)
- Predictor Variable - Checking Account Balance

Chi-square Test of Independence

- Chi-square test of independence starts with an assumption that there is no relationship between two variables.
- For example, we assume that there is no relationship between checking account balance and response.
    - H_0: There is no relationship between checking account balance and response. (The variables X and Y are independent.)
    - H_A: There is a relationship between checking account balance and response. (The variables X and Y are dependent.)

Motivation: Retail Sales

- Retailers can customize the online shopping experience by learning more about its customers. For example, Amazon wants to know if income level affects what shoppers look for (camera or phone) when they visit electronics.
- Use a chi-squared test for independence to answer this question.

Contingency Table

Purchase Category vs. Household Income (555 visitors to Amazon)

		Household Income					
		Less than $25,000	$25,000 to $49,999	$50,000 to $74,999	$75,000 to $99,999	$100,000 or more	Total
Purchase Category	Camera	26	38	76	57	50	247
	Phone	72	76	73	34	53	308
	Total	98	114	149	91	103	555

- Does the contingency table suggests that income and choice are dependent.

Observations from Contingency Table

- Association is evident suggesting that income and choice of product are dependent.
- Households with lower incomes seem more likely to purchase a phone; those with higher incomes a camera.
- Are these differences in purchase rates the result of sampling variation?

Chi-Squared test of independence

- Tests strength of association
- Tests the independence of two categorical variables using counts in a contingency table.

Chi-squared Tests

- Like other tests, the chi-squared test of independence relies on a test statistic-namely χ^2-to determine a *p*-value.
- We use this *p*-value as in other tests: we reject H_0 if the *p*-value is less than a chosen level of significance a (typically $\alpha = 0.05$).
- Chi-squared tests make assumptions, such as that the data are a simple random sample from the population of interest.

Hypotheses for the chi-squared test

- H_0: Household Income and Purchase Category are independent.
- H_1: Household Income and Purchase Category are not independent.

Calculating χ^2

- Measures the distance between the observed contingency table and a hypothetical contingency table.
- The hypothetical contingency table obeys H_0 while being consistent with observed marginal counts.

Calculating χ^2

		Household Income					
		Less than \$25,000	\$25,000 to \$49,999	\$50,000 to \$74,999	\$75,000 to \$99,999	\$100,000 or more	Total
Purchase Category	Camera						247
	Mobile phone						308
	Total	98	114	149	91	103	555

The null hypothesis determines expected cell counts in the hypothetical table.

Calculating χ^2

		Household Income					
		Less than \$25,000	\$25,000 to \$49,999	\$50,000 to \$74,999	\$75,000 to \$99,999	\$100,000 or more	Total
Purchase Category	Camera	43.61	50.74	66.31	40.50	45.84	247
	Mobile phone	54.39	63.26	82.69	50.50	57.16	308
	Total	98	114	149	91	103	555

The null hypothesis determines expected cell counts in the hypothetical table.

Calculating χ^2

$$\chi^2 = \sum \frac{(\textit{observed} - \textit{expected})^2}{\textit{expected}}$$

Accumulates the deviations between the observed and expected counts (in the hypothetical table) across all cells.

Calculating χ^2

		Household Income					
		Less than $25,000	$25,000 to $49,999	$50,000 to $74,999	$75,000 to $99,999	$100,000 or more	Total
Purchase Category	Camera	26	38	76	57	50	247
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$$\chi^2 = \frac{(26 - 43.61)^2}{43.61} + \ldots + \frac{(53 - 57.16)^2}{57.16} \approx 33.925$$

For retail data on purchase category and household income, the chi-squared statistic is 33.925 with 4 df. Since p value is $\leq \alpha = 0.05$, we reject the null hypothesis.

CHAID Tree Development

Following steps are used in developing a CHAID tree:

1. Root node.
2. Check the statistical significance of each independent variable depending on the type of dependent variable
3. The variable with the least p-value, based on the statistical tests described in step 2, is used for splitting the data set thereby creating subsets (represented by the internal nodes in the tree).
4. Using independent variables, repeat step 3 for each of the subsets of the data.
5. Generate business rules for the terminal nodes (nodes without any branches) of the tree.

CHAID Tree Development: German Credit Data

- Using a sample of 800 observations (file name: German Credit Rating.Xlsx) from the German Credit Data (can be downloaded from the University of California, Irvine machine learning data repository) to illustrate CHAID tree development.
- At each stage of the tree portioning, an appropriate hypothesis test has to be conducted for the variable selection.
- The null and alternative hypotheses are given below:
  - H_0: The variables Y and X are independent
  - H_A: The variables Y and X are dependent

Contingency table for German Credit

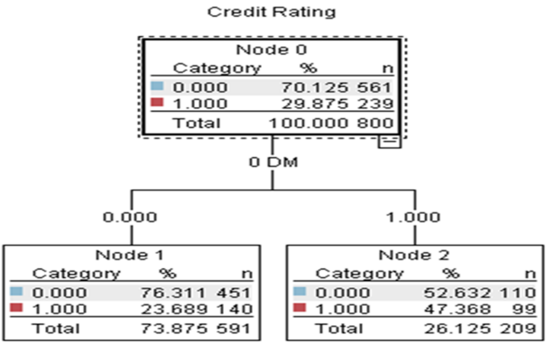
Checking Account Balance	Credit Rating (Observed)			Credit Rating (Expected)	
	Y = 1	Y = 0	Total	Y = 1	Y = 0
0DM = 1	99	110	209	62.44	146.56
0DM = 0	140	451	591	176.56	414.44
Total	239	561	800	239	561

Contingency table: Inference

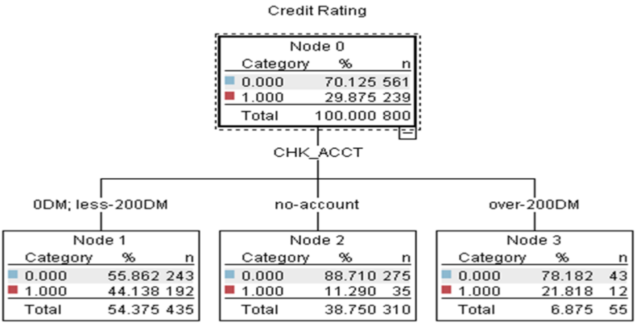
- Using the values in contingency table the Chi-square statistic is given by: $\chi^2 = 41.322$
- The chi-square critical value is 3.841.
- Since the chi-square statistic is much greater than chi-square critical value, we reject the null hypothesis.
- The corresponding *p*-value is 1.29E-10. That is, there is a statistically significant relationship between credit classification (Y) and checking account balance $0DM$.

CHAID-First Split

The CHAID tree split for the sample German credit data using checking account balance as the predictor variable is



CHAID Tree with all categories within checking account balance



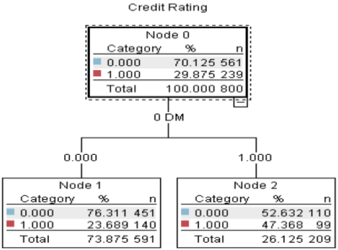
CHAID Tree with all categories within checking account balance

- The model has merged two checking account balance categories (0 DM and less than 200 DM) to create node 1. The pair of (predictor) categories that is least significantly different with respect to the dependent variable
- Node 2 represents the customers who do not have a checking account
- Node 3 represents those customers with more than 200 DM in their checking account balance.

Generating Business Rules using CHAID Tree

One of the advantages of decision tress is their use in generating business rules that can be deployed for decision-making.

CHAID-First Split

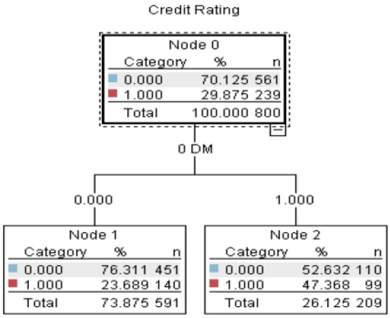


CHAID Tree

	OBSERVED		
Predictor	Response variable		Total
	Y = 1	Y = 0	
0DM = 1	99	110	209
0DM = 0	140	451	591
Total	239	561	800

	Expected		
Predictor	Response Variable		Total
	Y = 1	Y = 0	
0DM = 1	62.43875	146.56	209
0DM = 0	176.5613	414.44	591
Total	239	561	800

CHAID- Business Rules



CHAID Tree

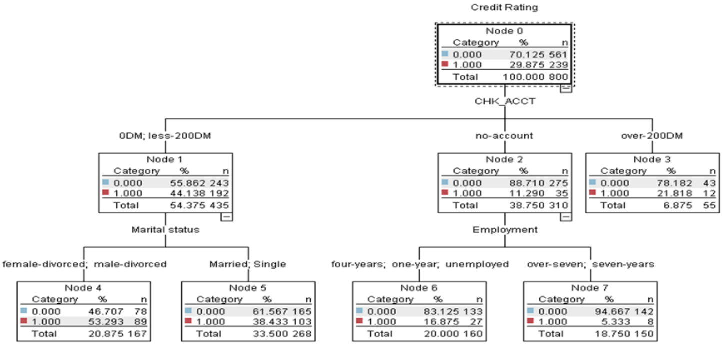
- Business rules are always derived for leaf nodes (node without any branch)
- Node 1: If 0 DM = 0 then classify the observation as 0 (good credit) which will be accurate 76% of the times

Building a tree: Key points

- Chi-square tests are applied at each of the stages in building the CHAID tree, to ensure that each branch is associated with a statistically significant predictor of the response variable.
- Bonferroni corrections, or similar adjustments, are used to account for the multiple testing that takes place.

CHAID including more Predictors

CHAID tree developed by using predictor variables such as checking account balance, marital status, and employment



Business Rules

- Nodes 3 to 7 are leaf nodes (terminal nodes) since there are no other branches emerging out of these nodes.
- Each terminal node is classified based on the largest class in that node.
  - For example, in node 3, the proportion of $Y = 0$ (good credit) is 78.182% and proportion of bad credit ($Y = 1$) is 21.818%.
  - Since the majority class in node 3 is 0, all the data in this node will be classified as 0 using the business rule checking account balance over 200 DM.
  - That is, when checking account balance is over 200 DM then the value of the outcome variable Y is 0.

Business Rules

Node	Business Rule	Support
3	Checking account balance is more than 200DM, classify the outcome as Y = 0. Classification accuracy is 78.18%.	6.875%
4	Checking account balance is either 0DM or less than 200DM AND the marital status is male divorced or female divorced, classify outcome as Y = 1. Classification accuracy is 53.293%.	20.875%
5	Checking account balance is either 0DM or less than 200DM AND the marital status is married or single, classify outcome as Y = 0. Classification accuracy is 61.56%.	33.500%
6	No checking account AND the employment is 1. Unemployed or 2. One year or 3. Four years Then classify the outcome as Y = 0. Classification accuracy is 83.125%.	20.00%
7	No checking account AND the employment is either seven years or over seven years then classify the outcome as Y = 0. The classification accuracy is 94.667%.	18.750%

For practical application, one may look for the business rules that have high accuracy as well as high support.

Classification and Regression Trees

- Classification and Regression Tree (CART) is a common terminology that is used for a Classification Tree (used when the dependent variable is discrete) and a Regression Tree (used when the dependent variable is continuous).
- Classification tree uses various impurity measures such as the Gini Impurity Index and Entropy to split the nodes.
- Regression Tree, on the other hand, splits the node that minimizes the Sum of Squared Errors
- CART is a binary tree, whereas CHAID can split the node into more than two branches.

Steps to Construct a Tree

The following steps are used to generate a classification and a regression tree (Breiman et al. 1984)

1. Step 1 : Start with the complete training data in the root node.
2. Step 2 : Decide on the measure of impurity (usually Gini impurity index or Entropy). Choose a predictor variable that minimizes the impurity when the parent node is split into children nodes.
3. Step 3 : Repeat step 2 for each subset of the data (for each internal node) using the independent variables until:
   - All the dependent variables are exhausted .
   - The stopping criteria is met.
4. Step 4 : Generate business rules for the leaf (terminal) nodes of the tree.

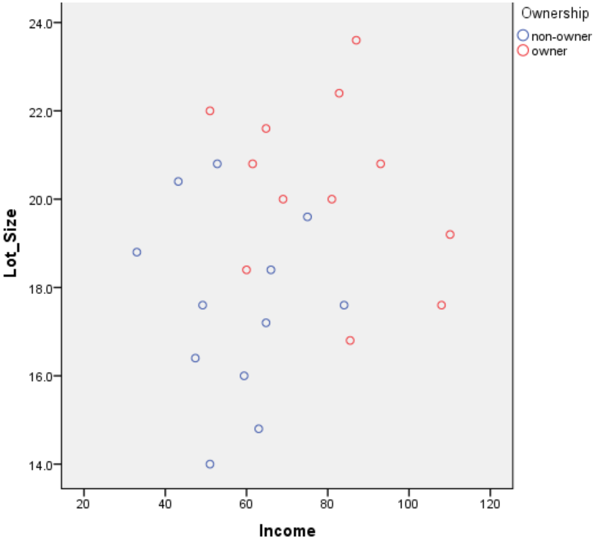
Example: Lawn Mowers

- A manufacturer of riding lawn mowers wishes to target an ad campaign to families who are most likely to purchase one.
- Is income and size of lot enough to discriminate owners and non-owners of riding lawn mowers?
- A sample of size $n_1 = 12$ current owners and $n_2 = 12$ current non-owners was obtained and X_1 = income and X_2 = size of lot were measured for each.

Data

π_1: Riding-mower owners		π_2: Nonowners	
x_1 (Income in \$1000s)	x_2 (Lot size in 1000 ft^2)	x_1 (Income in \$1000s)	x_2 (Lot size in 1000 ft^2)
90.0	18.4	105.0	19.6
115.5	16.8	82.8	20.8
94.8	21.6	94.8	17.2
91.5	20.8	73.2	20.4
117.0	23.6	114.0	17.6
140.1	19.2	79.2	17.6
138.0	17.6	89.4	16.0
112.8	22.4	96.0	18.4
99.0	20.0	77.4	16.4
123.0	20.8	63.0	18.8
81.0	22.0	81.0	14.0
111.0	20.0	93.0	14.8

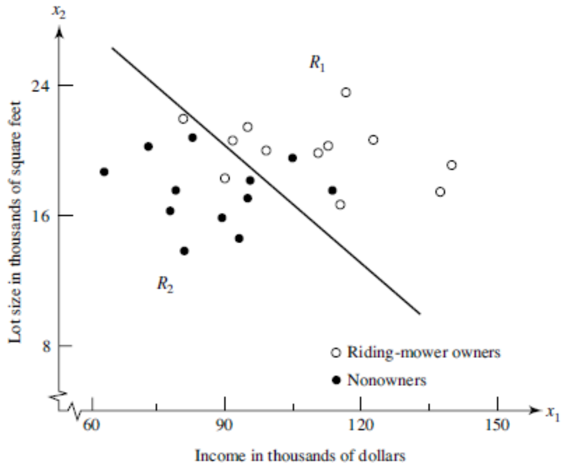
Scatter Plot



Observations

- Owners tend to have higher income and larger lots, but income appears to be a better discriminator.
- Note that the two populations, separated by the line, are not perfectly discriminated, so some classification errors are likely to occur.

Scatter Plot



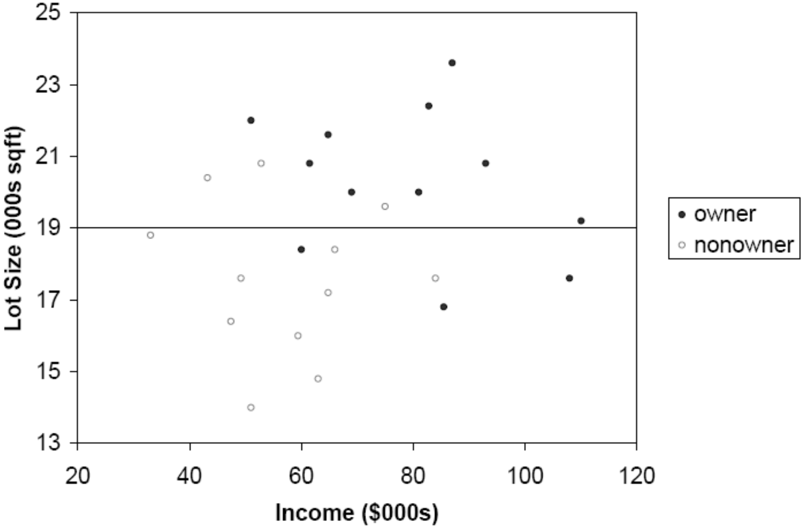
Recursive Partitioning

1. Pick one of the predictor variables, x_i
2. Pick a value of x_i, say s_i, that divides the training data into two (not necessarily equal) portions
3. Measure how "pure" or homogeneous each of the resulting portions are
   - "Pure" = containing records of mostly one class
4. Algorithm tries different values of x_i, and s_i to maximize purity in initial split
5. After you get a "maximum purity" split, repeat the process for a second split, and so on

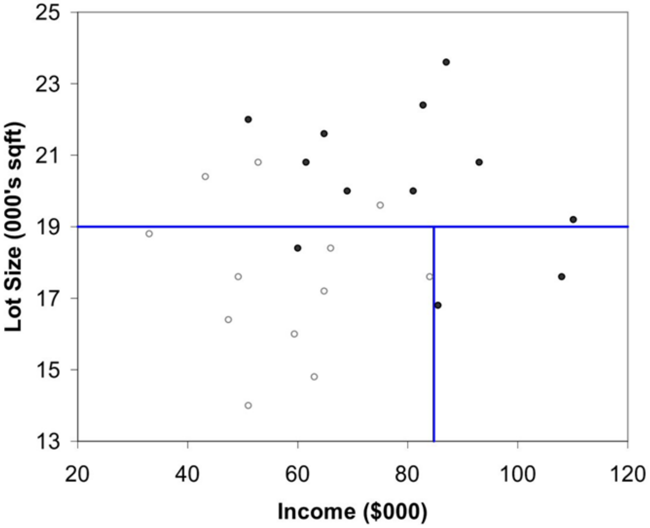
How to Split

- Order records according to one variable, say lot size
- Find midpoints between successive values E.g. first midpoint is 14.4 (halfway between 14.0 and 14.8)
- Divide records into those with lotsize > 14.4 and those < 14.4
- After evaluating that split, try the next one, which is 15.4 (halfway between 14.8 and 16.0)

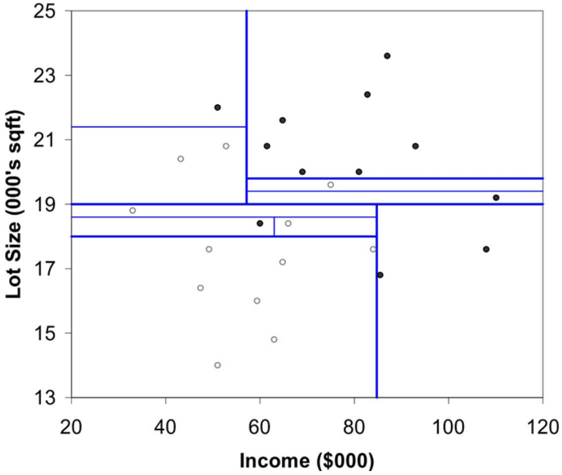
First Split: Lot Size=19,000



Second Split: Income=$84,000



After all Splits



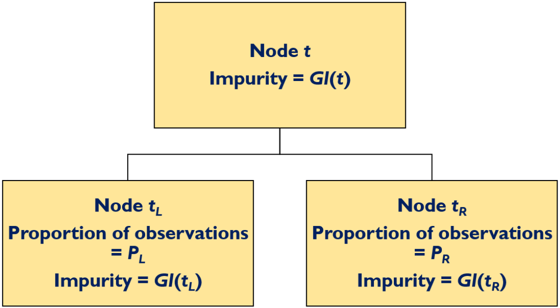
Gini Index

- A classification tree chooses the independent variable that minimizes the Gini impurity index.

$$GI(t) = \sum_{i=1}^{K} \sum_{j=1, j \neq i}^{K} P(C_i|t)P(C_j|t) = 1 - \sum_{i-1}^{k} [P(C_i|t)]^2 = 1 - 0.04 - 00$$

  - $GI(t)$ = Gini index at node t
  - $P(C_i|t)$= Proportion of observations belonging to C_i in node t
- For a classification problem with 2 classes, the Gini index value will lie between 0 and 0.5.
  - Higher value of the Gini index indicates higher impurity.
- At each step of the classification tree, the algorithm chooses the variable that provides the maximum reduction in Gini impurity.

Splitting strategy in CART



CART chooses a predictor variable, X_i, for splitting at node t that maximizes the function in

$$Max_{x_i}[GI(t) - P_L GI(t_L) - P_R GI(t_R)]$$

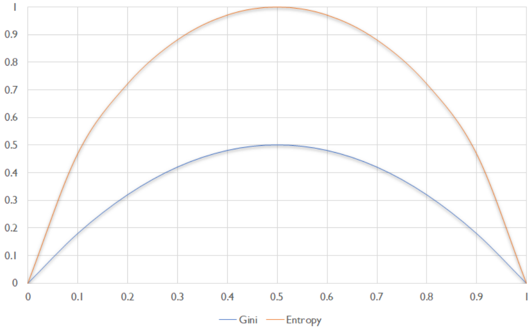
Entropy

- Entropy is another popular measure of impurity that is used in classification trees to split a node. It is given by

$$Entropy(t) = -\sum_{i=1}^{K} P(C_i|t) \times log_2 P(C_i|t)$$

- The value of entropy lies between 0 and 1, with a higher entropy indicating a higher impurity at the node.
- The value of entropy is higher than the value of Gini impurity index for a given proportion of positives and negatives.

Comparison of Gini index and Entropy



Cost Based Splitting Criteria

- Other than impurity measures such as Gini impurity index and entropy, decision makers may also use Cost of Misclassification to split the data.
- The total penalty is $C_{01}P_{01} + C_{10}P_{10}$ where
  - P_{01} = Proportion of 0 classified as 1
  - P_{10} = Proportion of 1 classified as 0
  - C_{01} = Penalty for classifying 0 as 1
  - C_{10}= Penalty for classifying 1 as 0

Regression tree

Regression trees are used to predict continuous outcome variable. The following steps are used for developing regression tree.

1. Start with the complete training data in the root node.
2. Split the root node by selecting a feature that gives the maximum reduction in sum of squared errors (SSE).

$$Max_x(SSE(t) - SSE(t_L, X_t) - SSE(t_R, X_t))$$

   is used for selecting the feature to split the root node to create the internal nodes.
3. Repeat Step 2 for internal nodes until it reaches a stopping criterion such as maximum depth from the root node.

Ensemble Methods

- The ensemble method is a machine-learning-algorithm that generates several classifiers (a classifier means a classification model) using different sampling strategies such as bootstrap aggregating.
- Random Forest is one of the popular ensemble method in which several trees (thus the name “forest”) are developed using different sampling strategies. One of the most frequently used sampling strategy is the Bootstrap Aggregating (or Bagging).
- Bagging is a random sampling with replacement

Summary

1. Decision trees are mostly used for solving classification problems and such trees are called classification trees.
2. They are developed using different splitting, merging, and stopping criteria.
3. There are several decision tree learning algorithms and they differ mainly in the splitting criteria.
   3.1 Splitting strategies Chi-square automatic interaction detection (CHAID) uses
      - chi-square test of independence when the dependent variable is categorical
      - F-test when the dependent variable is continuous
      - likelihood ratio test when the dependent variable is ordinal

   3.2 In classification and regression tree (CART) impurity measures such as
      - Gini impurity index or entropy are used as splitting criteria when the dependent variable is categorical
      - sum of squared errors (SSE) is used when the dependent variable is continuous.
4. One of the major advantages of decision tree learning algorithms is that it can be used for generating business rules from the model directly.